

Formulation and Scale Verification of "Effective Output" Based on Information Thermodynamics

— Mathematical Anatomy of Organizational Cognitive Limits, the 5 Phases of Complexity, and Phase Transitions —

Author: Hokuto Koyama (SBCM Alliance)

Contact: hokuto.kym@gmail.com (<mailto:hokuto.kym@gmail.com>)

ORCID: 0009-0001-7903-0100

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Abstract

This study elucidates the "productivity paradox"—where despite the elimination of information generation costs due to advancements in AI technology, maintenance and coordination costs in modern organizations increase non-linearly, leading to stagnant productivity—from the perspectives of information thermodynamics and linear algebra. We redefine labor and decision-making as energy conversion processes and formulate a "complexity penalty" in dimensions of watts (W) based on Landauer's principle and the human biological metabolic limit (20W). In response to the explosive increase in input information volume (N), we derive five thermodynamic phase transition phases through which a system progresses from optimal equilibrium, through a "parasitic dissipative structure," to functional cessation. We mathematically prove that the macroscopic optimal organizational scale follows $P_{opt} \propto N^{2/3}$. Furthermore, in the micro-geometric section, individual skills are redefined as projection operators from high-dimensional tensors to low-dimensional manifolds. We formulate the explosive increase in dissipation heat caused by the "phase difference (orthogonality)" between the unconstrained output of AI and the highly localized cognitive manifolds of experts, proving the paradox that more highly skilled experts exceed their metabolic limits earlier, triggering a cascade of organizational collapse. This paper presents the

geometric requirements of orthogonalization of information structures and prior dimensionality reduction as the sole conditions for saving the system from thermodynamic collapse (vacuum decay).

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1. Introduction: The Physical Reality of "Cognitive Load" in the AI Era

1.1 Problem Statement: Disappearance of Generation Costs and Non-linear Increase in Maintenance Costs

The explosive advancement of artificial intelligence (AI), particularly generative AI technologies such as large language models, in the early 21st century has triggered a historical, irreversible paradigm shift in human intellectual production activities. This is the fact that the marginal cost of "information generation (Execution)" has approached zero. However, despite this dramatic technological benefit, a subjective sense of fatigue and management rigidity are pervasive in modern organizations, and the "modern productivity paradox" (Brynjolfsson et al., 2017) is worsening, where productivity improvements are neither statistically nor experientially observed.

This study argues that the root of this paradox lies not in the "generation" of information, but in its "grounding." Massive amounts of information generated at high speed by AI do not function in the real world as they are. Human mutual verification and coordination processes are inevitable to bind them seamlessly with physical

constraints, logical consistency, and existing organizational systems. As a result of the disappearance of generation costs, the output information elements (N) increase, and the "maintenance and coordination costs" to eliminate contradictions among these elements and ensure the consistency of the entire system are increasing non-linearly. Traditional economic indicators (GDP and working hours) fail to capture this "invisible explosion of management and coordination costs," leaving organizations exhausted for unknown reasons and on the verge of self-destruction.

1.2 Paradigm Shift: From Working Hours (Scalar) to Energy Conversion (Thermodynamics)

To elucidate the true nature of this non-linear organizational exhaustion, this study completely discards the measurement paradigm based on 1-dimensional scalar quantities such as "working hours" and "production volume," upon which traditional economics and business administration have relied. Instead, human cognitive and labor activities are defined as "physical work required to reduce information uncertainty (entropy)," and the organization is strictly redefined as an information-thermodynamic energy conversion process (heat engine).

Information is not an abstract concept but has physical substance. The process of determining (fixing) a single state from a group of uncertain information and erasing noise is an irreversible physical computation. Therefore, the act of humans giving meaning to information, resolving contradictions, and making organizational decisions can be described as purely thermodynamic phenomena. By converting labor from "time" to the dimension of "power (Watts: $[J/s = W])$ " consumed and emitted by living organisms, it becomes possible to place organizational limits on the foundation of physical laws.

1.3 Objective: Quantifying the Complexity Penalty and Deriving Phase Transition Limits

The primary objective of this study is to quantify the invisible management and coordination costs plaguing modern organizations as "physical exhaust heat (Joules)"—that is, the "complexity penalty"—inevitably generated when eliminating information uncertainty.

By formulating the thermodynamic load brought about by the increase in the information generation rate (N) as a fundamental equation, and introducing the absolute hardware limit (metabolic limit) of the human brain as a boundary condition, we mathematically derive five thermodynamic phase transition phases. These phases

range from an "optimal state," through a "parasitic dissipative structure" that temporarily obfuscates local entropy increases, to an ultimate "vacuum decay" resulting in functional cessation. This paper is an attempt to detach the sustainability of human organizations in the AI society from ambiguous discourses in social sciences and prove it as an absolute ruling based on physical laws.

2. Cross-sectional Integration of Prior Studies (Literature Review)

The theoretical framework of this study horizontally integrates findings on the "penalties and limits associated with increased complexity," which have developed independently in the fields of information theory, thermodynamics, neuroscience, and complexity science, under a set of fundamental equations.

2.1 Metabolic Constraints in Information Thermodynamics and Neuroscience (Landauer & Laughlin)

The starting point of the physical axioms in this model is Landauer's principle (1961), which posits that "information is physical, and its erasure inevitably entails heat dissipation of $k_B T \ln 2$." Friston's (2010) free-energy principle demonstrated that the brain's inference and decision-making process itself is a thermodynamic process attempting to minimize surprise (information entropy) between its internal model and the external environment. Furthermore, neuroscientific studies by Laughlin et al. (1998) quantitatively show that the metabolic limit (hard cap) the human brain can allocate to information processing is only about 20W. This study introduces this "exhaust heat accompanying the erasure of uncertainty" and the "biological 20W constraint" into the model as absolute physical hard caps for the entire macroscopic organizational system.

2.2 Diminishing Marginal Returns in Complexity Science and History (Tainter)

Regarding the mechanisms of system complexity increase and collapse, the historian J. Tainter's (1988) *The Collapse of Complex Societies* is highly suggestive. Tainter proved the historical structure where, when societies or organizations increase their internal complexity to solve new problems, they initially gain marginal returns; however, eventually, the energy and costs required to maintain that complexity exceed the returns, leading to collapse (regression to a lower-dimensional, simpler structure).

This study mathematically substantiates this macroscopic historical and sociological observation as a process of thermodynamic energy balance failure accompanying information processing.

2.3 Overfitting Penalties in Information Theory and Statistics (AIC/BIC)

The concept that an increase in information volume does not always contribute to value creation for a system aligns with the structure of the "mathematical penalty against the number of variables (complexity)" seen in the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) in statistics. If parameters (information) are increased to enhance the descriptive power of a model, the goodness of fit improves, but the price for maintaining the internal consistency of the entire system (the risk of overfitting and computational cost) increases as a penalty. This study reinterprets the concept of this mathematical penalty as a "thermodynamic price" in information processing systems.

2.4 Combinatorial Explosion in Institutional Economics and Organizational Networks (Coase & Metcalfe)

From the perspective of organizational theory, Coase's (1937) transaction cost theory explained the boundary between markets and organizations in terms of management costs. In addition, Brooks's (1975) *The Mythical Man-Month* in software engineering and Metcalfe's law in communication networks indicate that communication and coordination paths for N nodes undergo a combinatorial explosion at $O(N^2)$. Recently, the proliferation of "bullshit jobs" (meaningless management and coordination tasks) pointed out by Graeber (2018) from a sociological approach is often discussed as a malfunction of capitalism. However, according to the framework of this study, it is not a sociological phenomenon but an "accumulation process of entropy generation accompanying a combinatorial explosion" that is structurally inevitable in networks, and it can be explained as a dissipative phenomenon with physical inevitability.

[Part 1] Macro-Thermodynamics: Information Explosion as a Scalar and Biological Limits

3. Theoretical Assumptions, Defensive Lines, and Falsifiability (Limitation & Falsifiability)

To establish this model not merely as a metaphor or a sociological concept but as a rigorous, scientific, and falsifiable theory, we explicitly state the three physical and mathematical preconditions upon which this study relies, as well as the boundaries to which the model is applied (intentional abstractions and defensive lines).

3.1 Equivalence of Information Entropy and Thermodynamic Entropy

[Assumption] Invoking Friston's (2010) free-energy principle, this study defines that "the 'rejection of uncertainty (decision-making and consistency checking)' in the human brain is a physical, irreversible process." The erasure of information follows Landauer's principle (1961) and inevitably entails exhaust heat (entropy generation) proportional to $k_B T \ln 2$ (k_B is the Boltzmann constant, T is the absolute temperature) per bit at room temperature.

[Boundary and Defensive Line] What this model measures is not the subjective "fatigue" or "stress" reported by workers. It is the direct measurement of entropy generation (thermodynamic dissipation) inevitably caused by physical laws when the nervous system determines a single state from multiple uncertain states (resolving logical contradictions). Psychological factors are outside the scope of this model.

3.2 Macro-constraint of the Biological Inefficiency Coefficient (β)

[Assumption] There is a massive scale difference between the theoretical Landauer limit ($k_B T \ln 2 \approx 2.87 \times 10^{-21} \text{J/bit}$) and the metabolic limit exhibited by actual biological brains (approx. 20W), representing the gap between microscopic phenomena and macroscopic organs. In this study, the inevitable entropy increase associated with the active transport of neurotransmitters, the opening and closing of ion channels, and background maintenance metabolism when the brain executes logical operations is collectively introduced as a macro "biological inefficiency coefficient ($\beta \gg 1$)" functioning as a heat engine.

[Boundary and Defensive Line] This study does not delve into elucidating the microscopic neurophysiological breakdown of β (e.g., which cell groups consume ATP and how). Neuroscientific mechanisms are black-boxed and treated purely as a macroscopic, thermodynamic "constant constraint condition" of the entire system.

3.3 Assumption of a Fully Connected Network (Worst-case Scenario)

[Assumption] It is assumed that information coordination costs within an organization increase at $O(N^2)$ relative to the number of nodes N . This presupposes a "fully connected state (worst-case scenario)" where task and authority dependencies are undifferentiated, and all output nodes must mutually verify consistency to ensure there are no contradictions.

[Falsifiability] Conversely, in an ideal environment where task dependencies are completely decoupled (orthogonalization of information is achieved), or in a system where information modularization functions perfectly, the $O(N^2)$ explosion of exhaust heat predicted by this model does not occur. The conditional validity of this premise is precisely what guarantees the primary falsifiability of this theory.

4. The "Three-Phase Composition" of Working Hours and Formulation of Effective Output (I-EO)

Based on the physical assumptions in the previous chapter, the human labor process is redefined as an energy conversion process (heat engine) of an information processing system, and its dimensions are uniformly formulated into power (Watts: $[J/s = W]$).

4.1 Physical Decomposition of the Labor Phase

From the perspective of information processing, the total working hours of a worker (T_{total}) are divided into the following three phases, and the processing information volume (I) for each is defined.

- **Work Phase (T_w):** The phase that directly creates physical and logical value (e.g., writing code, making decisions, physical operations). This corresponds to the pure work the system performs on the outside.
- **Monitoring Phase (T_{mo}):** The phase that verifies the individual validity of the output (number of elements N) generated by automated systems such as AI. The processing information volume here is linearly proportional to the number of elements N ($I_{mo} = c_1 N$, where c_1 is a proportionality constant).
- **Maintenance and Coordination Phase (T_{ma}):** The phase that resolves contradictions among output elements and ensures the consistency of the entire system. Since mutual confirmation of all combinations among elements is required, the processing information volume increases with the square of the number of elements ($I_{ma} = c_2 \frac{N(N-1)}{2} \approx \frac{c_2}{2} N^2$, where c_2 is a constant).

4.2 Fundamental Equation: Derivation of Information-Thermodynamic Effective Output (I-EO)

Based on the above, the net productivity of a system (individual or organization) is newly defined as "Information-Thermodynamic Effective Output (I-EO)." This is the net power calculated by subtracting the complexity penalty (exhaust heat requirement) expended to eliminate uncertainty from the gross power generated by the system.

$$I-EO = \frac{P_{work} \cdot T_w - \overbrace{\beta \cdot k_B T \ln 2 \cdot (I_{mo} + I_{ma})}^{Q_{penalty}}}{T_{total}} \quad [W]$$

- **Physical Meaning:** The generated information (N) itself is not value. The true power of the system (I-EO) is what remains after subtracting the process ($Q_{penalty}$) of "grounding" that information group to physical reality and mutual logical constraints to reduce the entropy known as internal contradiction. As N explosively increases due to AI, the processing information volume of the monitoring and coordination phases (especially $I_{ma} \propto N^2$) becomes dominant, causing $Q_{penalty}$ to expand non-linearly.

4.3 The 20W Constraint as a Hardware Limit

Due to biological constraints, the upper limit of energy that the brain can constantly allocate to information processing (rejection of uncertainty) based on the human body's metabolic capacity is physically fixed at $P_{max} \approx 20W$.

As the information volume N increases due to the improved generation capacity of AI, the complexity penalty that must be processed per unit of time ($\frac{Q_{penalty}}{T_{total}}$), i.e., the exhaust heat requirement, rapidly increases. When this required heat dissipation approaches or reaches the hardware limit P_{max} , the system becomes physically unable to execute any further "rejection of uncertainty (decision-making and consistency checking)." This state does not mean a mere "delay in operations" but a functional failure as a heat engine, namely, the Thermal Death of the system.

##[Part 1] Macro-Thermodynamics: The 5 Phase Transitions of an Organization and Macro-scale Verification

5. Phase Transition Model: The "5 Thermodynamic Phases" of an Organization

In the effective output equation ($I-EO$) defined in the previous chapter, as the input information volume N (the number of elements output from generative systems such as AI) increases, the quadratic increase ($O(N^2)$) of the complexity penalty non-linearly changes the net power of the system (organization and individual). Through this dynamic process of change, the system undergoes thermodynamic phase transitions, transitioning through the following five distinct phases.

5.1 Phase 1 (Under-load Phase)

The region where N is sufficiently small and $I-EO > 0$. This is a state where the input information volume is insufficient relative to the system's information processing capacity. Although the operational efficiency of the entire system is low, the exhaust heat requirement (complexity penalty $Q_{penalty}$) accompanying the resolution of contradictions among elements and the erasure of uncertainty is also minimal, making it a highly thermodynamically stable state.

5.2 Phase 2 (Optimum Equilibrium Phase)

The point where the external information generation rate (input N) and the processing rate at which the system grounds that information to physical and logical reality to reduce entropy are thermodynamically optimally balanced, maximizing (peaking) $I-EO$. This indicates the ideal state where the net power of the system is at its highest, and the organization is most capable of converting energy into its original purpose of "creating value for the outside."

5.3 Phase 3 (Critical Point ①: Red Queen Limit / N_{RQ}^*)

As N further increases, the consistency maintenance cost (I_{ma}), which scales at $O(N^2)$, begins to dominate the system's energy balance, asymptotic to $I-EO \rightarrow 0$. The entire energy of the system is spent solely on "resolving internal contradictions and eliminating the uncertainty of the increased information group," and the pure work output to the outside theoretically ceases. This is a thermodynamic critical point equivalent to the "Red Queen Hypothesis" in evolutionary biology, where the system must run at full speed just to maintain its internal state.

5.4 Phase 4 (Parasitic Dissipative Structure)

An anomalous region where $I-EO < 0$ (negative effective output). Normally, the system should collapse at this point due to a deficit in its energy balance; however, as a characteristic of non-equilibrium open systems, a "secondary structure to process the generated uncertainty (heat) itself (e.g., the proliferation of coordination specialists, consultants, and middle management)" self-organizes within the system.

[Explanation by the Law of Conservation of Energy] This life-prolonging measure by the dissipative structure does not arise from nothing. It barely constructs a local reduction in entropy by borrowing in advance against continuous external energy injection (burning investment capital) or the reserve metabolic limits of the individuals comprising the system (overwork, exploitation of mental resources). This is the physical reality of the state of "being busy but producing nothing" that many modern organizations have fallen into.

5.5 Phase 5 (Critical Point ②: Vacuum Decay Phase / N_{burn}^*)

Figure 1. Thermodynamic Phase Transition Diagram of Effective Output

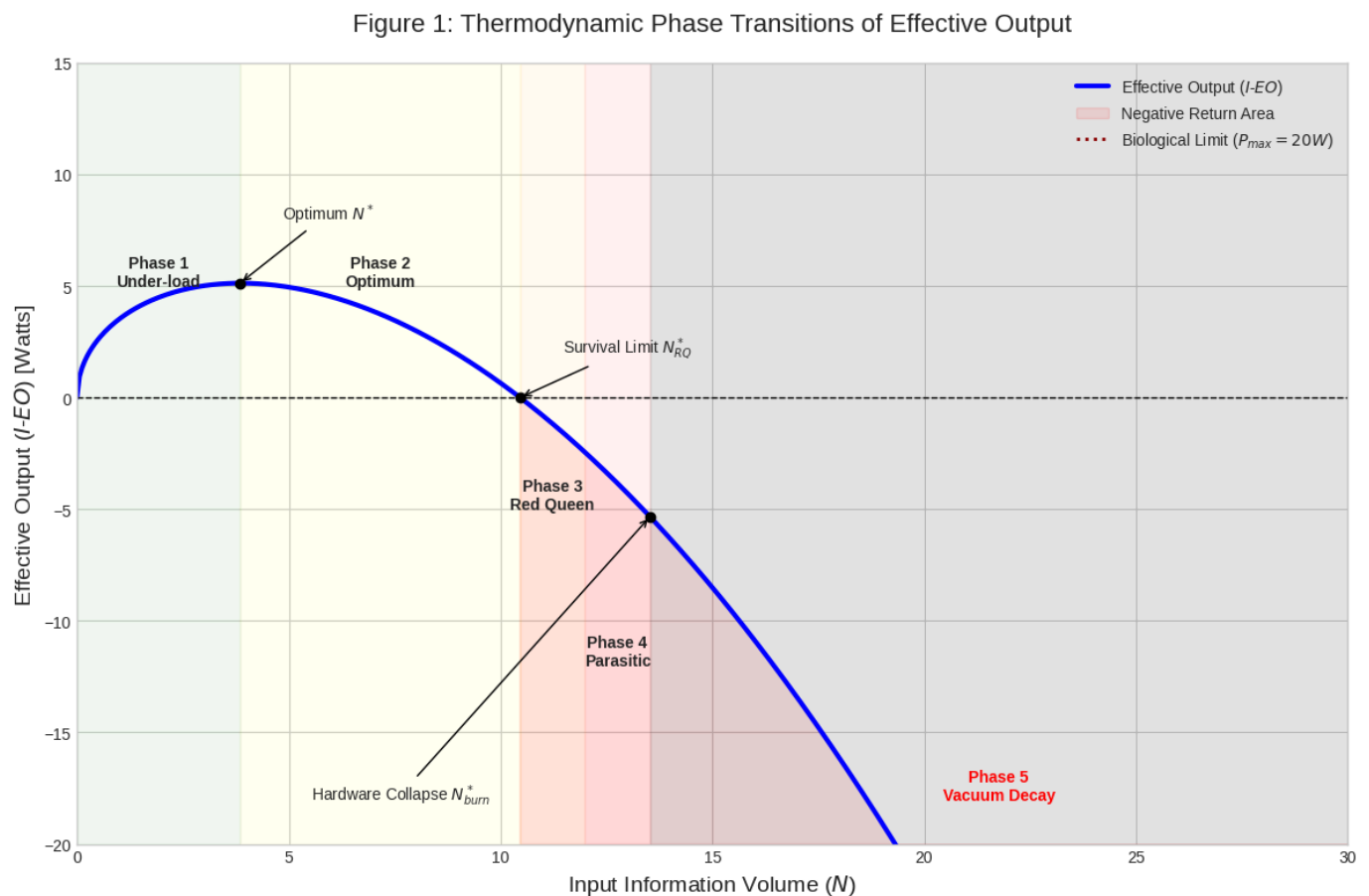


Figure 1 visualizes the changes in the effective output ($I-EO$) of the system as the input information volume N increases. Starting from Phase 1, where the information volume is small, and passing through Phase 2 (Optimum Equilibrium Phase), the system irreversibly enters Phase 3 (Red Queen Phase). What is particularly noteworthy

is the behavior from Phase 4 onward. The cost of checking the consistency of the massive amount of information generated by AI begins to demand the metabolic limit (20W) of the brain, and the effective output sharply plummets into the negative. At the critical point of N_{burn}^* , the system can no longer exhaust heat and reaches a "vacuum decay (Phase 5)," which is a structural self-destruction.

This is the point where the information explosion continues, and the total exhaust heat requirement of the system completely exceeds the maximum heat dissipation capacity of the biological brain ($P_{max} \approx 20W$), which is the hardware limit. This critical point N_{burn}^* is proportional to the square root of the hardware limit.

$$N_{burn}^* \propto \sqrt{P_{max}}$$

At this point, the external energy that can be borrowed in advance and the reserve metabolism of the individuals are completely depleted, making the macroscopic structural maintenance of the system (organization and individuals) physically impossible. It is an irreversible reset phenomenon of the system, observed as the burnout of constituent elements or a complete functional cessation of organizational decision-making.

6. Macro-scale Verification: Derivation of Optimal Organizational Scale

P_{opt}

This section proves that when the micro-equations based on individual biological limits from the previous chapters are scaled up to a macroscopic organization composed of a large number of individuals, this theory maintains mathematical consistency without diverging or collapsing.

6.1 Formulation of the Macro Total Exhaust Heat Function $\delta(P, N)$

The total exhaust heat (total entropy generation) $\delta(P, N)$ that an entire organization with a population of P individuals must process is defined as the sum of the following two terms.

- **System-facing Exhaust Heat (Information Sharing):**

When an information group of N elements is evenly divided among P individuals, the processing elements per individual become N/P . Since the exhaust heat required for checking internal consistency borne by each individual is proportional to the square of the elements, it becomes $\propto (N/P)^2$. The sum of this for the

entire organization (for P individuals) is the term $\alpha P(N/P)^2 = \alpha \frac{N^2}{P}$ (where α is the heat conversion coefficient).

- **Human-facing Exhaust Heat (Organizational Friction):**

However, if information is shared among individuals, communication paths are then required for those P individuals to mutually transmit and synchronize information as a system. According to Metcalfe's law, coordination entropy among members increases in proportion to P^2 , so this term becomes γP^2 (where γ is the communication and coordination inefficiency coefficient).

- **Total Exhaust Heat Function:**

The sum of these two conflicting exhaust heat requirements becomes the organization's total exhaust heat function.

$$\delta(P, N) = \alpha \frac{N^2}{P} + \gamma P^2$$

6.2 Derivation of Local Minimum and Proof of Scale Validity

When a fixed information volume N is given to the system, we seek the thermodynamically optimal organizational scale P that minimizes the entropy generation (total exhaust heat) $\delta(P, N)$ of the entire system. By partially differentiating this function with respect to P , we solve for the local minimum ($\frac{\partial \delta}{\partial P} = 0$).

$$\frac{\partial \delta}{\partial P} = -\alpha \frac{N^2}{P^2} + 2\gamma P = 0$$

$$2\gamma P = \alpha \frac{N^2}{P^2}$$

$$P^3 = \frac{\alpha}{2\gamma} N^2$$

$$P_{opt} = \left(\frac{\alpha}{2\gamma} \right)^{1/3} N^{2/3}$$

$$\therefore P_{opt} \propto N^{2/3}$$

6.3 Conclusion: The Physical Inevitability of Modularization

This mathematical derivation completely negates the naive linear scaling hypothesis that "simply increasing personnel proportionally will solve the problem ($P \propto N$)" when the information volume N increases.

In response to the information load on the system, an organization can only expand its scale at a pace of $N^{2/3}$. If personnel are injected beyond this, the frictional heat among the organization's members (γP^2) will accelerate the complexity penalty, driving the system to an early Phase 5 (self-destruction). In other words, this theory derives a rigorous macroscopic conclusion that "for any given information volume N , there is always exactly one unique optimal scale that is thermodynamically most stable." It proves that the reason a massive system must be divided (modularized) is not due to ambiguous psychological reasons such as the "limits of human management capacity," but rather originates from "physical and thermodynamic exhaust heat limits."

Figure 2. Thermodynamic Optimization of Organizational Scale: Proof of $P_{opt} \propto N^{2/3}$

Figure 2: Thermodynamic Optimization of Organization Scale

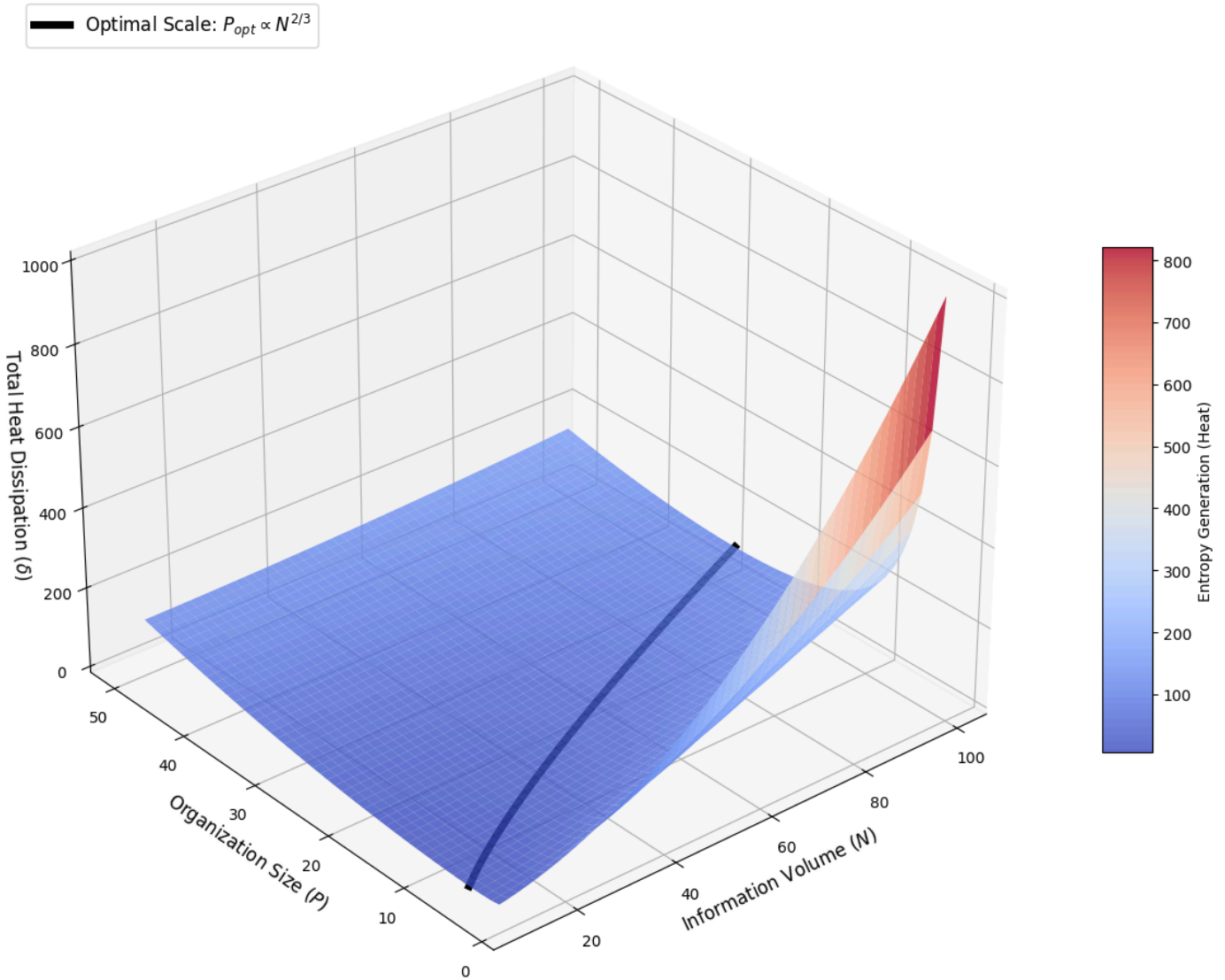


Figure 2 shows the organization's total exhaust heat function $\delta(P, N)$ in 3D. The blue "valley bottom" running through the center of the graph indicates the thermodynamically optimal organizational scale P_{opt} , where the total exhaust heat is minimized relative to the information volume N . The trajectory of this valley bottom follows the mathematically derived $P_{opt} \propto N^{2/3}$. If the organization deviates from this valley and increases personnel P too much, a red cliff caused by "coordination frictional heat (Metcalf's law)" rises. Conversely, if there are too few personnel, thermal runaway occurs due to "individual overload." This figure proves the physical inevitability that an organization's scale expansion cannot be linear ($P \propto N$), and under an information explosion, the organization must be finely divided (modularized).

[Part 2] Micro-Geometry: Projection Limits of Cognitive Manifolds and "Orthogonality of Skills"

7. From Scalar to Vector: Geometrical Redefinition of "Skill"

In addition to the macroscopic metabolic limits against "information volume N (scalar)" proven in Part 1, this chapter formulates the microscopic and geometrical structure of the complexity penalty caused by the "dimensions (vectors and tensors)" of information.

7.1 Limitations of the 1-Dimensional Scalar Model

Conventional labor economics and human resource management models have defined an individual's information processing capacity (skill) as a 1-dimensional scalar quantity, such as "processing speed" or "energy efficiency." However, there are strict physical and anatomical upper limits to the number of dimensions of variables that the human nervous system can simultaneously hold and manipulate (the capacity of working memory and the extent of cognitive schemas). In modern labor, which builds and maintains complex systems, the input information is not merely a continuous scalar but forms a multi-dimensional vector space of interdependent variables. Therefore, the conventional approach of treating capability as a scalar is incapable of describing the true nature of the cognitive load brought about by AI.

7.2 Projection Operator H_i onto a Low-dimensional Manifold

This study linearly and algebraically redefines the "skill (cognitive characteristics)" of an individual i not as the absolute speed of processing in the information space, but as a linear mapping from a high-dimensional information space to a "low-dimensional manifold" that the individual can process—that is, a "Projection Operator H_i ." The act of a human verifying the validity of information and eliminating (determining) uncertainty is, geometrically, nothing other than "an operation of projecting a high-dimensional input tensor onto one's own cognizable low-dimensional manifold."

7.3 Deflection (Specialization) of Eigenspaces and the Reality of Expertise

The cognitive characteristics (projection operator H_i) possessed by any individual i do not constitute an Identity Matrix that acts isotropically on all dimensions. It is described as an asymmetric projection matrix that has high eigenvalues (or singular values) only in the direction of specific basis vectors, truncating the other dimensions to zero (rank-deficient).

This "extreme bias towards specific dimensions (lower dimensionality)" is precisely the mathematical reality of "expertise" and "skill" in humans, enabling the high-speed execution of specific tasks while keeping biological metabolism at a basal level.

8. Generation of "Dissipation Heat" due to Dimensional Alignment and Phase Differences

Based on the projection operator model defined in the previous chapter, we formulate the thermodynamic "dissipation" mechanism that occurs when high-dimensional tensors output from unconstrained information generation systems like AI are input into human limited cognitive manifolds.

8.1 High-dimensional Output Tensor $\mathbf{X}_{AI}$

Mechanical information generation systems such as AI are not subject to the biological metabolic limit of 20W or the constraints of working memory. Therefore, the output information group is generated as a high-dimensional tensor $\mathbf{X}_{AI}$ that simultaneously encompasses numerous mutually orthogonal dimensions, such as local logical structures, global system architectures, security requirements, and even past contexts.

8.2 Projection onto Orthogonal Complements and Tensor Formulation of Dissipation Heat

When an individual i attempts to verify (monitor and coordinate) the validity of this high-dimensional input $\mathbf{X}_{AI}$ by projecting it onto their own low-dimensional eigenspace to eliminate uncertainty, a fatal thermodynamic phenomenon occurs. Information components belonging to the "Orthogonal Complement" of the individual's eigenspace cannot be interpreted or processed on that individual's cognitive manifold, and no information compression or semantic determination (entropy reduction) takes place. This unprocessed entropy stagnates within the system as thermodynamic "Dissipation Heat ($Q_{dissipation}$)."

The magnitude of this dissipation heat is strictly formulated as the projection error onto the orthogonal complement (the square of the Frobenius norm), using the identity matrix I and the projection matrix H_i of individual i .

$$Q_{dissipation} \propto \|(I - H_i)\mathbf{X}_{AI}\|^2$$

What this formula signifies is that the further the dimensions of the input information deviate from the individual's cognitive manifold, the more the system is forced into futile logical operations "attempting to forcefully determine (erase) unknown

dimensions," thereby generating massive exhaust heat based on Landauer's limit.

8.3 Dynamic Functionalization of Phase Difference θ and Biological Inefficiency Coefficient $\beta(\theta)$

We define the angle (phase difference) formed by the principal component vector of the input tensor $\mathbf{X}_{AI}$ and the eigenspace of the individual as θ . At this time, the macroscopic biological inefficiency coefficient β introduced in Part 1 is no longer a constant but is redefined as a dynamic function $\beta(\theta)$ that fluctuates non-linearly depending on the phase difference θ .

$$\beta(\theta) = \frac{\kappa}{\cos \theta + \epsilon} \quad (\epsilon \ll 1, \kappa \text{ is the basal metabolic constant})$$

- $\cos \theta \rightarrow 1$ (**Parallel / Alignment Match**):

When the dimensional vector of the input information completely matches the individual's eigenspace, the information loss (norm) due to projection is minimized. The exhaust heat accompanying the elimination of uncertainty converges to a constant multiple of Landauer's limit (the basal metabolic level κ), and the system maintains a highly effective output ($I-EO$).

- $\cos \theta \rightarrow 0$ (**Orthogonal / Alignment Mismatch**):

When the major dimensions of the input information are orthogonal to the individual's eigenspace (i.e., they include dimensions that are completely outside their expertise or jurisdiction), the information volume on the individual's projection space asymptotically approaches zero ($H_i \mathbf{X}_{AI} \approx \mathbf{0}$). Since the total amount of entropy possessed by the input information stagnates in the system as dissipation heat, the denominator $\cos \theta$ approaches zero, and the inefficiency coefficient $\beta(\theta)$ diverges to an astronomical figure.

As a result, the effective output ($I-EO$) defined in Part 1 instantly plummets to an extreme negative value, and the individual instantaneously exceeds the hardware limit (20W).

9. Mathematical Proof of the Paradox: The Thermodynamic Vulnerability of "Brilliant Experts"

Why do brilliant experts, who boast extremely high information processing speed and accuracy (capabilities as scalar quantities) in specific domains, reach their metabolic limits (burnout) the fastest in response to the high-dimensional outputs brought about

by the introduction of AI? This counter-intuitive, paradoxical phenomenon occurring frequently in modern organizations is mathematically proven using the projection operator model from the previous chapter.

9.1 Linear Algebraic Definition of Expertise (Rank Deficiency and Localization of Bases)

"High-level expertise" in an information processing system does not mean the absolute "speed" of processing the information space indiscriminately. In terms of linear algebra, it refers to a state where the biological brain has acquired a projection matrix H_{expert} in which singular values are extremely concentrated only in the direction of a few specific basis vectors, completely truncating the other dimensions (an intentionally rank-deficient state).

Due to this extreme dimensionality reduction and the localization of the eigenspace, experts have maximally minimized the entropy generation of information erasure for specific inputs (parallel to their own domain of expertise), thereby maximizing their local effective output ($I-EO$). This is the physical and geometric structure of experts under the traditional division of labor system.

9.2 Forced Input of Orthogonal Dimensions and Maximization of Dissipation Heat

However, the omnidirectional, high-dimensional tensor $\mathbf{X}_{AI}$ output unconstrainedly by AI contains massive amounts of "basis components completely orthogonal" to the expert's highly localized eigenspace, such as the consistency of the overall architecture, dependencies with other domains, and transformations in business logic. When an expert is required in their duties to verify and coordinate the overall validity of this $\mathbf{X}_{AI}$, the components belonging to the orthogonal complement are maximized. Consequently, the norm of the expert's projection error $\|(I - H_{expert})\mathbf{X}_{AI}\|^2$ is maximized. Mathematically, an expert optimized by truncating dimensions exhibits a larger projection error against high-dimensional inputs than a non-expert (generalist) with a versatile (high-rank) projection matrix.

9.3 Theoretical Consequence: Exceeding Biological Hard Caps due to Phase Mismatch

Driven by a high sense of responsibility and professional ethics, experts attempt to forcibly determine (erase) the unknown orthogonal dimensions using their own low-dimensional eigenspace (specialized cognitive schema).

As a result, the phase-difference-dependent biological inefficiency coefficient $\beta(\theta)$ defined in the previous chapter increases exponentially, and even with a small amount

of information input, the total amount of dissipation heat rapidly exceeds the hardware limit of the brain (20W). This burnout phenomenon is not a decline in individual ability or negligence; it is an absolute thermodynamic consequence caused by a "geometrical dimensional mismatch" between the high-dimensionality of the input tensor and the low-dimensionality of the individual's projection manifold.

Figure 3. The Paradox of Experts: Geometrical Anatomy of Skill Orthogonality and Dissipation Heat

Figure 3: 3D Geometrical Anatomy of the Expertise Paradox

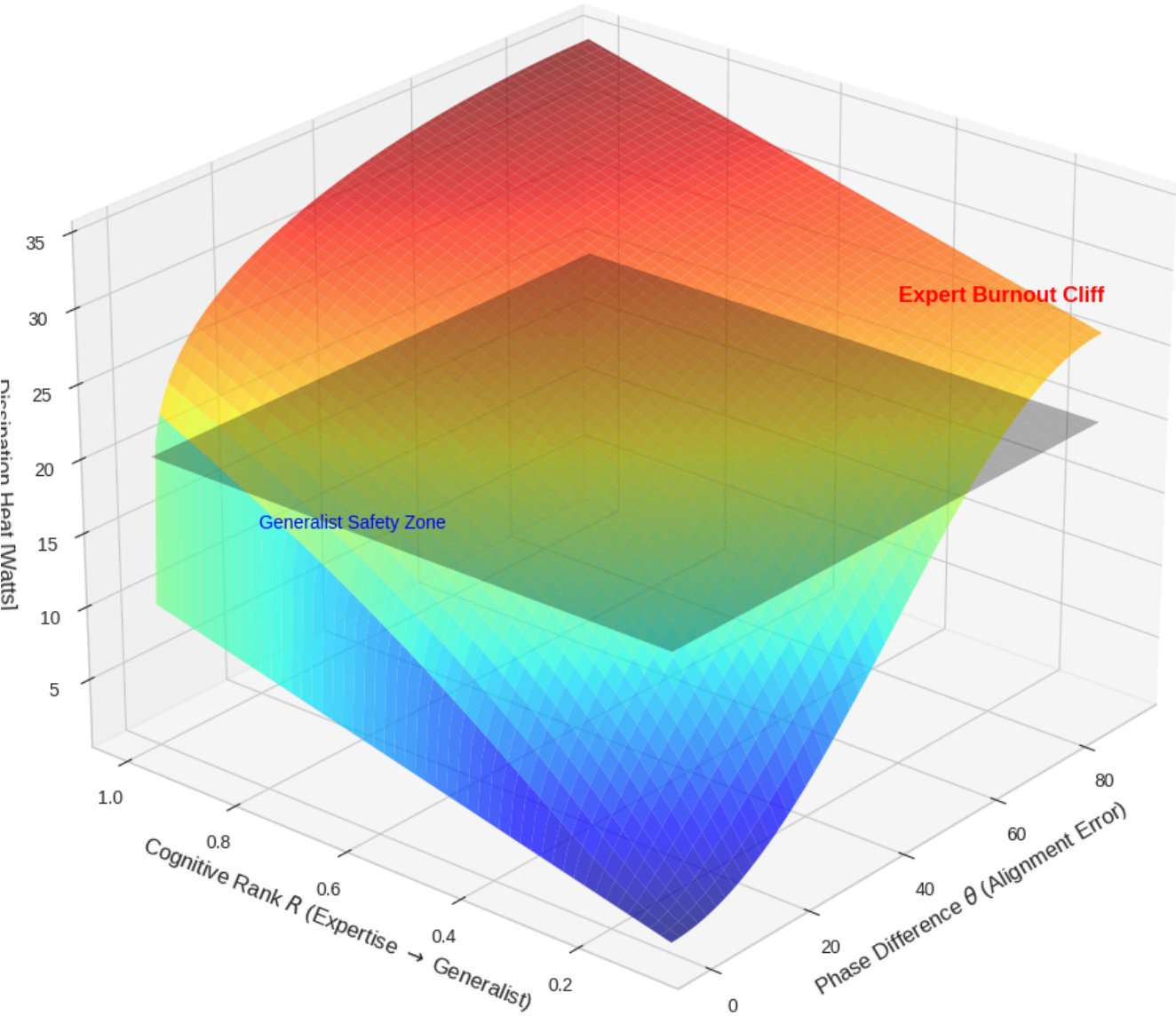


Figure 3 geometrically proves the "thermodynamic vulnerability of experts," which is the core of this paper. The horizontal axis, phase difference θ , represents the "orthogonality (deviation)" between the AI's output dimensions and the individual's cognitive manifold, while the vertical axis, Cognitive Rank R , represents the sharpness of expertise.

The "Expert Burnout Cliff" appearing in the right foreground of the figure shows that the more highly specialized an expert is (low rank R), the more explosively the dissipation heat increases even when a slight phase difference (θ) with the AI output occurs. Experts boast extremely high efficiency near their area of expertise ($\theta = 0$), but the moment they touch the "orthogonal components" possessed by AI-derived high-dimensional information, they vertically breach the 20W limit line (black plane) and self-destruct. In contrast, generalists (high rank R) maintain a wide survivable zone, albeit with lower efficiency. This figure geometrically highlights the paradox that in the AI era, more highly skilled experts are the first to meet a "thermodynamic death."

10. Cascading Collapse of Organizational Coordination (T_{ma}): Entropy Increase in Mutual Projections

How does the exceedance of the micro-individual metabolic limit (20W) trigger a macro phase transition of the entire organization (the transition from Phase 4 to Phase 5), rather than merely resulting in an individual's leave of absence? This chapter formulates the cascading failure process over the network.

10.1 Formulation of Mutual Projections and Translation Entropy

In the maintenance and coordination phase (T_{ma}) where individual i and individual j within an organization mutually verify and coordinate the consistency of the information output by AI, the bases (expertise and cognitive schemas) of their respective projection operators H_i and H_j do not perfectly match.

When information that has been processed (reduced in dimensionality) in the eigenspace of individual i is re-projected onto the eigenspace of individual j to form a consensus, costs of information loss and reinterpretation inevitably occur. This "translation entropy ΔS_{ij} " increases in proportion to the distance between the two projection matrices (for example, the Frobenius norm $\|H_i - H_j\|_F$).

10.2 Generation of Residual Noise due to Local Metabolic Exceedance ($I-EO < 0$)

Due to the increase in information volume N , the individual in the network with the largest "phase difference θ with the input information (such as the specific experts proven in the previous chapter)" is the first to reach the metabolic upper limit, resulting in $I-EO < 0$.

At this breaking point, the individual becomes completely unable to eliminate (determine) the uncertainty of the assigned information tensor. Unprocessed errors, contradictions, and undetermined variables stagnate on the organizational network as thermodynamic "residual noise (unprocessed entropy)" and spill over into the next process.

10.3 The Dynamics of Cascade Propagation (Avalanche of Phase Transitions)

When this residual noise flows into an operationally adjacent individual j , it acts as a mutated tensor $\mathbf{X}'_{AI}$ that further increases the "dimensionality" and "uncertainty" of the input tensor for individual j .

As a result, even for individual j , who should originally have remained within their processing limits, the orthogonal complement component $\|(I - H_j)\mathbf{X}'_{AI}\|^2$ is pushed up in a chain reaction, causing individual j 's exhaust heat requirement to also breach its limit. This phenomenon, where the "non-linear increase in one's own dissipation heat due to the unprocessed entropy of others" propagates sequentially to adjacent nodes, is the cascade propagation of organizational collapse.

This deterministic thermodynamic process is the true mathematical mechanism behind the organization's irreversible downfall from Phase 4 (local and temporary life extension through parasitic dissipative structures) to Phase 5 (structural buckling and vacuum decay of the entire system).

11. Bridge to Conclusion: Geometric Requirements for Orthogonalization of Information Structures and Dimensionality Reduction

Given the rigid physical constraints of the individual biological metabolic limit (20W) and the upper limit of the projection operator's rank (the number of dimensions that can be processed simultaneously), this chapter defines the geometric requirements of system design to prevent the system from reaching Phase 5 (Vacuum Decay Phase).

11.1 Physical and Geometric Denial of Cognitive Manifold Expansion (Limits of Reskilling)

The approach frequently advocated in modern management environments—namely, "retraining (reskilling) humans to understand and process all output dimensions of AI"—is completely denied within the framework of this theory.

Attempts to expand an individual's cognitive manifold into an omnidirectional, high-dimensional space are synonymous, in terms of linear algebra, with the operation of artificially increasing the rank of individual i 's projection operator H_i . However, expanding processing dimensions and acquiring versatility means the hypertrophy of information processing (the entropy reduction process) on the manifold, accompanied by a constant increase in dissipation heat based on Landauer's limit. This inevitably only accelerates the collision with the biological hardware upper limit (20W).

Therefore, absorbing complexity by expanding individual capabilities is a thermodynamically impossible approach.

11.2 Orthogonalization of System Topology and Prior Dimensionality Reduction

The only mathematical solution to save the system from thermodynamic collapse is not to expand human processing capacity, but to geometrically transform the information structure itself in the "pre-stage" before the input tensor $\mathbf{X}_{AI}$ is fed to the individual.

Specifically, the massive high-dimensional tensor generated by AI must be pre-decomposed into mutually independent low-dimensional basis vector groups (Orthogonal Decomposition). Furthermore, a "system-side topology design" that filters and routes information so that the phase difference between the eigenspace possessed by each individual i (a specific one-dimensional or low-dimensional manifold they excel in) and the input pre-divided vector becomes $\cos \theta \rightarrow 1$ (perfectly parallel) is an essential requirement.

Only when the modularization of information and the orthogonalization of dependency relations are achieved will the dissipation heat $\|(I - H_i)\mathbf{X}_{AI}\|^2$ asymptotically approach zero, enabling humans to continue maintaining a high effective output ($I-EO$) within their own eigenspace.

12. Conclusion: Absolute Rulings by Physical Laws and the Final State of the System

12.1 The True Identity of the Complexity Penalty

The reality of the "productivity paradox due to the introduction of AI" facing modern organizations is not a sociological or psychological phenomenon such as subjective "busyness," "management failure," or "decline in motivation."

Its true identity is the "**absolute thermodynamic entropy reduction cost (accumulation of Landauer's limit)**" required to determine the state of high-dimensional information groups, which have been freed from generation cost constraints, so that they conform to physical reality and logical constraints. This is observed as the sum of the "**dissipation heat due to phase difference**" accompanying the projection onto the low-dimensional manifolds of individuals, and the "**local dissipative structure of Phase 4 (meaningless coordination tasks)**" that proliferates to compensate for the system's energy balance deficit.

12.2 Transformation of Individual Roles: From Subjects of Value Creation to Thermodynamic Dissipation Organs

In a society where the marginal cost of information generation asymptotically approaches zero, the role of human individuals in the information processing system has undergone a decisive transformation. Humans are no longer the "generators" of information. Their physical function has degraded (or been purified) into an "**entropy exhaust organ utilizing 20W of biological metabolism**" designed to eliminate the uncertainty of high-dimensional information output unconstrainedly by AI and adapt it to physical reality.

In particular, a tragic and mathematical paradox has been proven here: experts with a sharp basis in a specific single dimension (low rank) generate the maximum dissipation heat against inputs of orthogonal dimensions, thereby exceeding their metabolic limits and falling into functional failure the earliest within the system.

12.3 Summary

The combinatorial explosion of networks ($O(N^2)$), the dissipation heat due to dimensional mismatches in projection manifolds, and the metabolic upper limit of the biological brain (20W). Injecting unconstrained AI output into an organization while ignoring these three physical and mathematical constraints inevitably leads the system to Phase 5 (macro structural buckling and vacuum decay due to exceeding the total exhaust heat requirement).

The formulation in this study completely proves, as a falsifiable mathematical model, that the sole necessary and sufficient conditions for a system (organization) composed of human elements to survive in the AI society are not ambiguous

spiritualism or management techniques, but rather a **"thermodynamically optimized scale limit of $P_{opt} \propto N^{2/3}$ "** and the **"perfect orthogonal decomposition of information tensors."**

Data and Code Availability

All mathematical models and simulation codes presented in this study are publicly available in the following GitHub repository. This includes the Python notebook files (.ipynb) used to generate Figures 1, 2, and 3 of this paper.

- **Repository URL:**https://github.com/SBCM-Alliance/core-theory/tree/main/Complexity_Penalty (https://github.com/SBCM-Alliance/core-theory/tree/main/Complexity_Penalty)

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